



Department of Information Technology

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch : IV Semester-B. Tech Information Technology

Course Code & Title : IT3401 Web Essentials

Name of the Faculty member : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 1 / Web Browsers and Web Servers
- **Course Outcome** : CO1
- **Topic Learning Outcome:** TLO7
- **Activity Chosen** : Mind Map
- **Justification:**
 - A mind map activity is a graphical representation of ideas and concepts, serving as an effective visual thinking tool. This activity has been chosen to enhance the understanding of web browsers and web servers.
 - Through this activity, students will gain a clear understanding of key concepts and the functionalities of both web browsers and web servers.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - Instructor explained the functionalities of web browser and web server with examples.
 - Based on the discussion and after clarifying the student's doubts, the faculty asked the students to draw a mind map related to the topic within 15 minutes.
 - Students individually created mind maps on Web Browsers and Web Servers, using visual representations to illustrate the concepts covered in class. These sheets were subsequently collected by the instructor.
 - This activity improves students grasp of the topic and makes it easier for them to respond to questions, as illustrated in Fig.2.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO1	3	2	2	1	1	1

(1 – Low 2 – Moderate 3 – High)



• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	Students will be able to understand the functionalities of web browser and web server	Students will be able to demonstrate the capabilities of web browser and web server	Students will be able to illustrate the interactions between web browser and web server in graphical representation	As the student involved in the activity as an individual, the individual thinking skill improved	Students will be able to communicate technical concepts effectively	Students will be able to understand the functionalities of web browsers and web servers, and relate them to computational problem-solving by applying mathematical and computing skills

• Images / Screenshot of the practice:



Fig 1: Mind Map Activity

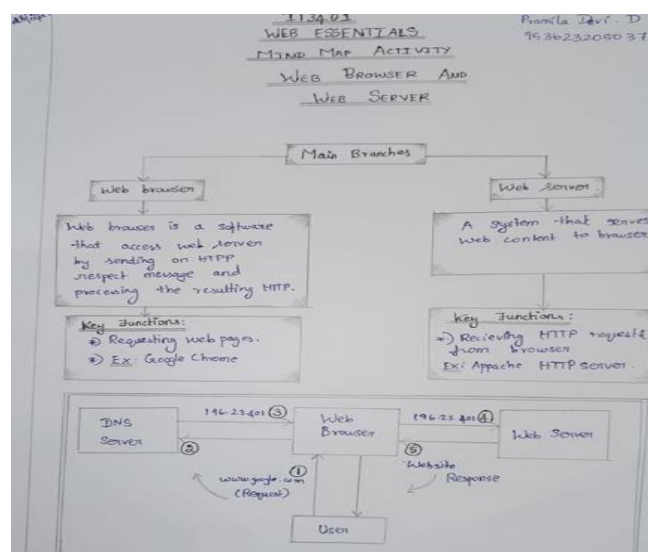


Fig 2: Sample Sheet of Mind Map Activity of Student D. Pramila Devi



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Students expressed that creating a mind map significantly helped them to visualize the communication process between browsers and servers.
- Students felt that through this activity, it became easier to remember key concepts such as HTTP methods and server responses.
- They felt comfortable to ask doubts with their peers.

- ❖ **Benefit of the practice:**

- The mind map activity proved to be an effective teaching strategy in enhancing student engagement and understanding.
- It allowed students to actively organize and connect key concepts related to web browser and web server functionalities and their communication process.
- This activity encouraged students to be more interactive and collaborate effectively with their peers
- Through this activity, students were able to achieve better clarity and understanding of the topic, particularly in terms of how web browsers and web servers function and communicate.

- ❖ **Challenges faced in implementation:**

- Some students struggled to recall or differentiate between key technical terms such as HTTP, DNS, client, server, and protocols.
- The activity took longer for some students to complete, as they needed additional time to understand the concepts and organize their thoughts effectively.

The Student S.Anitha represented the Mind Map activity on Web browser and web server communication with following mistakes

- i) "She did not include the role of the DNS server in the communication process between the web browser and the web server."
- ii) She omitted the representation of HTTP request and response messages in the mind map diagram."

In the next class, I guided the students on how to visually connect all the key functionalities of the web browser and server, and also instructed them to present their mind maps in a clear and organized manner.

References:

- https://www.ritrjpm.ac.in/images/computer-science/5_CS8591_Mindmap.pdf

Signature of Faculty Member

HOD